

Project Design Phase-II Technology Stack (Architecture & Stack)

Date	25-06-2025
Team ID	23A51A05G1
Project Name	Metro Ticket Generating System in ServiceNow
Maximum Marks	4 Marks

Technical Architecture:

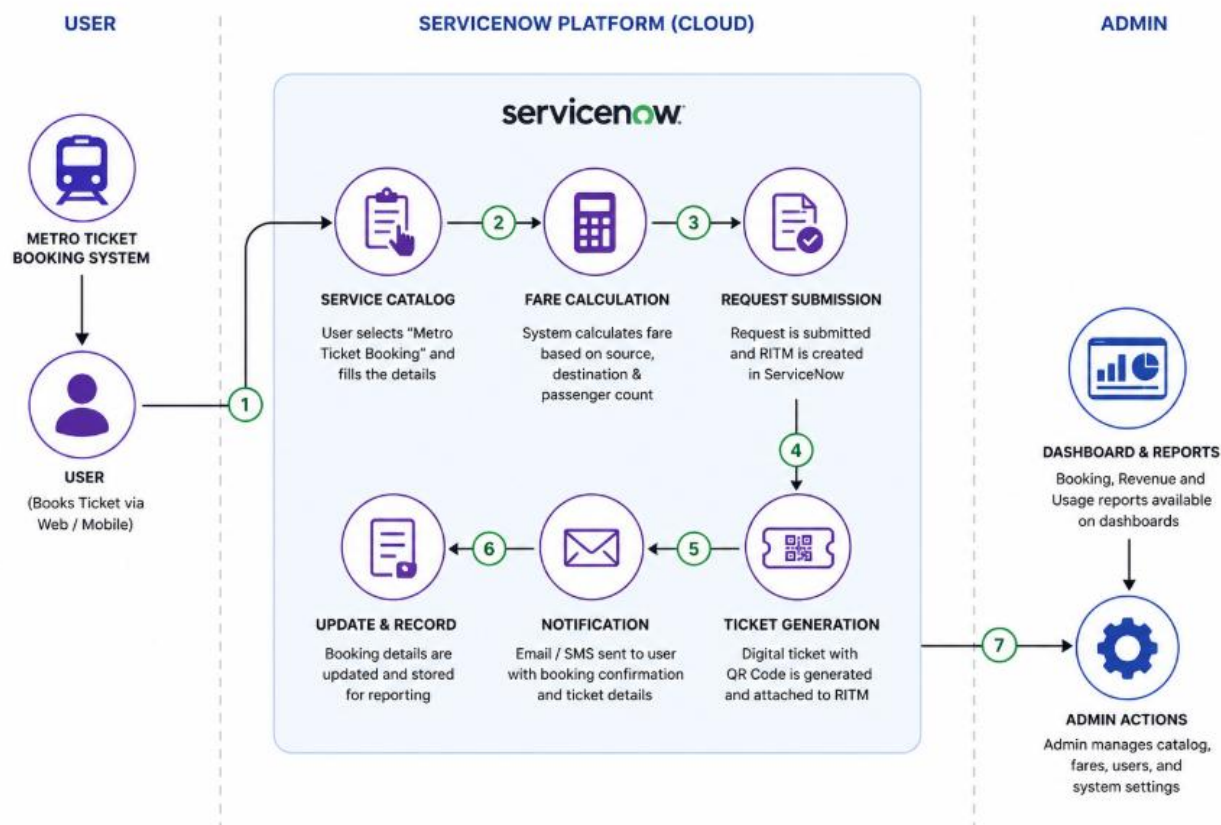


Table-1 : Components & Technologies:

S.No	Component	Description	Technology
1.	User Interface	Interface for passengers to book metro tickets	ServiceNow Service Catalog, Portal UI
2.	Application Logic-1	Fare calculation based on source and destination stations	Catalog Client Scripts (JavaScript)
3.	Application Logic-2	Server-side fare retrieval and validation	Script Includes, GlideAjax
4.	Application Logic-3	Automated ticket booking workflow and notifications	Flow Designer, Business Rules
5.	Database	Stores metro stations, fares, and booking information	ServiceNow Tables (u_metro_station, u_metro_fare, sc_req_item)
6.	Cloud Database	Cloud-based data storage managed by ServiceNow	ServiceNow Cloud Database
7.	File Storage	Stores QR codes, ticket attachments, and records	ServiceNow Attachments
8.	External API-1	QR Code generation for digital tickets	QR Code Generator API / Scripted QR Generation
9.	External API-2	Email notification service for ticket confirmation	ServiceNow Notification Engine
10.	Machine Learning Model	Ticket booking trends, revenue, and usage reports	ServiceNow Reports & Dashboards
11.	Infrastructure (Server / Cloud)	Application hosting and execution environment	ServiceNow SaaS Cloud Platform

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1.	Open-Source Frameworks	Client-side scripting and customization used for fare calculation and form validation	JavaScript, GlideAjax
2.	Security Implementations	Role-based access control for administrators and users, authenticated access through ServiceNow platform, secure data storage and transmission	ServiceNow ACLs, User Roles, HTTPS, IAM Controls
3.	Scalable Architecture	ServiceNow's cloud-based multi-tenant architecture supports growing users and ticket bookings without infrastructure changes	ServiceNow SaaS Architecture, Multi-Tenant Cloud Platform
4.	Availability	High availability through ServiceNow cloud infrastructure with continuous monitoring and managed services	ServiceNow Cloud Infrastructure, Redundant Data Centers
5.	Performance	Fast ticket booking process through server-side processing, optimized database queries, and cloud-hosted services	GlideAjax, Script Includes, ServiceNow Database Optimization

References:

<https://www.servicenow.com/docs/r/servicenow-platform/service-catalog/service-catalog.html>

<https://www.servicenow.com/docs/r/application-development/building-applications/flow-designer.html>

https://www.servicenow.com/docs/r/api-reference/c_GlideAjaxAPI.html

<https://developer.servicenow.com>

<https://www.servicenow.com/community/>